Kubernetes CI/CD Pipeline
Six Practical Questions

Instructions for all questions

- Implement the application in VS Code using Python.
- Use the following five-stage pipeline: (1) Source Code, (2) Version Control, (3) Continuous Integration, (4) Build, and (5) Deployment.
- In Continuous Integration, GitHub Actions must check out the code, install dependencies, and run automated tests.
- After tests pass, build a Docker image and push it to Docker Hub.
- Deploy the image to Kubernetes and expose the application through a Service.
- Make exactly two source-code pushes from VS Code to GitHub. Push 1 must trigger the first deployment; after modifying the application, Push 2 must trigger a new pipeline run and deploy the updated version.
- Use a unique Docker image tag for each push. The deployment must use the image built from that push.

Practical Questions

1. Menu-Driven Calculator

Develop a Python application with menu options for addition, subtraction, multiplication, and division.
Push 1: Implement addition and subtraction.
Push 2: Add multiplication and division.

2. Student Grade Calculator

Develop an application that accepts student marks and displays the total, average, and grade.
Push 1: Calculate total and average marks.
Push 2: Add grade calculation based on the average.

3. Shopping Bill Calculator

Develop a menu-driven billing application that accepts an item's price and quantity.
Push 1: Calculate the total bill.
Push 2: Add a discount calculation.

4. Temperature Converter

Develop an application with menu options to convert Celsius to Fahrenheit and Fahrenheit to Celsius.
Push 1: Implement Celsius-to-Fahrenheit conversion.
Push 2: Add Fahrenheit-to-Celsius conversion.

5. Simple Banking Application

Develop a menu-driven application with options to check balance, deposit money, and withdraw money.
Push 1: Implement balance checking and deposits.
Push 2: Add withdrawal functionality with balance validation.

6. Library Book Management

Develop a simple application with menu options to add a book, view available books, and issue a book.
Push 1: Implement adding and viewing books.

Push 2: Add the book-issuing option and update book availability.